Torrin Leonard

Product Engineer, Full-Stack & AI

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Canadian citizen. Remote only, anywhere in Canada. Not seeking on-site or hybrid.

Summary

Product engineer, full stack and applied AI. I build and own production products end to end with React, TypeScript, Node and Python, and I take AI features from 0 to 1: retrieval-augmented generation (RAG) and vector search, fine-tuning, and guards against prompt injection and hallucination. I ran production releases and mentored developers at Equator, and ran my own studio before that. I work through coding agents (Claude Code) daily: I plan first, test every change, and keep the agent's work reviewable, so a mistake is caught before it ships and not repeated.

Experience

Founder & Engineer, Builderloo

Feb 2026 - Present | Self-employed / Waterloo, ON

Tech: TypeScript, Node.js, aweft, Postgres, cheerio, MCP, OAuth 2.1, Claude Code, Linux, nginx

- Built and run Builderloo.ca alone, a community platform for Waterloo Region tech (formerly KWBuilds): a directory of 851 local startups (October 2026), their open roles, local events and news, a forum, and a weekly newsletter. 507 commits.
- Scrapers read 4 startup hubs, 18 news sources and 5 event calendars, plus every listed company's careers page and blog, on one 1 GB server.
- Chose browserless scraping (got, cheerio) after a measured bake-off across 493 sites recovered content for 98.8%, retiring the headless browser and Docker from the stack.
- Serves a public MCP server with a hand-rolled OAuth 2.1 authorization server (Dynamic Client Registration, PKCE) so claude.ai and ChatGPT connectors can attach while anonymous clients still work.
- Build it with Claude Code agents from written work orders, with tests and reviews on every change.

Full Stack Software Developer, Equator Studios

Mar 2023 - Sep 2026 | Hybrid / Waterloo, ON

Tech: Node.js/Express, React, Next.js, TypeScript, MongoDB, Python/FastAPI, OpenAI, Qdrant, ChromaDB, GeoPandas, Docker, GitLab CI/CD, AWS, Linux

- At a seven-person startup, built a 0 to 1 AI proposal product as its sole engineer through the initial build, then led it as its development team grew to three, authoring the majority of commits and 64% of the code.
- Rebuilt the marketing site on Next.js and owned it end to end through CI/CD and release, growing organic search from under 1,000 clicks and under 1M impressions to 52.1K clicks and 6.17M impressions in six months.
- Authored and merged 70 production releases on the AI product, and served as release manager for a main-product release, coordinating the rollout through pre- and post-push testing.
- Built the RAG ingestion and retrieval pipeline on OpenAI embeddings and Qdrant, migrated off ChromaDB, using query-expansion retrieval rather than single-vector search.
- Implemented a client model fine-tuning pipeline on the OpenAI fine-tuning API, and a proxy metering token cost per user on every model call.
- Built an admin prompt orchestration layer so non-engineers tune production AI behaviour live, with versioned code-defined fallbacks and prompt-injection and hallucination guards.
- Integrated server-hosted GIS segmentation models (SamGeo, LangSAM) for satellite imagery selection in the main app.
- Added Stripe payment systems and supported site-based pricing rollout with product.
- Onboarded and mentored 2 full-time developers, reviewed 55 merge requests, and set written acceptance criteria on work handed off to them.
- Created and decomposed 2 company epics, sat on interview panels, and held the architecture decision on the AI product and the frontend.
- Wrote and maintained the team developer handbook, commit and merge-request policy, and engineering principles with a self-review checklist.

Co-Founder & CEO, Lead Software Developer, This Cozy Studio

Oct 2021 - Mar 2023 | Waterloo, ON

Torrin Leonard, Product Engineer, Full-Stack & AI https://torrin.me

Tech: React, Node.js, Django, Python, Blender API, HTML/CSS/JavaScript, AWS

- Founded and ran a federally incorporated 3D modeling and rendering contract pipeline studio across 7 client engagements with no outside funding; owned product, clients, contracts, and delivery end to end.
- Built per-client combinatorial rule engines and unattended batch rendering on rented GPU capacity, delivering over 50,000 rendered files to one client in under a week.
- Open-sourced the studio's Blender tooling (see Projects) and grew a companion tutorial channel to 282,388 views and 3.29K subscribers across 25 videos, driving adoption.

Automation & Accessibility Engineer (Contract / Part-time), League

Mar 2021 - May 2022 | Remote

Tech: JavaScript/TypeScript, TestCafe, Node/npm, Git, Jira, WCAG

- Built TestCafe regression suites in TypeScript/JavaScript, reducing manual cycles and improving release confidence.
- Performed WCAG accessibility audits and partnered with engineers/PMs to ship compliant releases; triaged defects within CI workflows.

Software QA Tester (Contract / Part-time), worX4you

Jun 2013 - Mar 2021 | Contract / Various Clients

- Tested web and mobile apps part time, on and off, for clients including TunnelBear, Hubba and Hopscotch: manual, cross-browser and regression testing.

Projects

maccobylab.com | https://maccobylab.com

No-code web platform for behavioral research: researchers build an experiment visually, share one link, and get charts and raw CSV back.

- Compiles each experiment to a constrained JSON definition, validates and freezes it server-side against a generated op whitelist, then runs it in participants' browsers on a purpose-built interpreter rather than generated code.
- Runs author-written code in an opaque-origin sandboxed iframe with no network access, and measures reaction times on a monotonic high-resolution clock so wall-clock changes cannot corrupt them.

MCP dashboard

Sandboxed execution platform for agent-authored modules, running live.

- Agent-authored modules stored as Postgres rows, each run in an isolated child process that cannot reach the filesystem or its peers.
- An egress proxy injects credentials outside module code, so agent-written modules never hold a plaintext secret.

aweft | https://aweft.dev

Full-stack TypeScript toolkit for apps whose state is an observable document, published to npm under MIT as @aweftjs.

- Built solo across 399 commits: 23 packages covering state, sync, storage, server, UI, sandboxing, jobs, and auth. Builderloo, maccobylab.com, my MCP dashboard, and torrin.me all run on it.

Blender generative rendering add-on | https://github.com/torrinworx/Blend_My_NFTs

Blender add-on automating procedural 3D asset generation and export, written against the Python/Blender API.

- 1,015 stars and 324 forks, used by multiple studios. Authored 423 of 496 commits across 7 releases.

Skills

Languages: JavaScript, TypeScript, Python, Rust
Frontend: React, Next.js, destamatic-ui, HTML/CSS
Backend: Node.js, Express, FastAPI, Django, REST APIs, WebSockets
Databases: MongoDB, PostgreSQL, SQL, Qdrant, ChromaDB
AI & Data: OpenAI API (embeddings + fine-tuning), RAG (Retrieval-Augmented Generation), vector search, evals, prompt injection defense, Whisper, pandas
AI-assisted development: Claude Code and OpenCode daily: plan first, test every change, keep agent work reviewable, build systems that fail safely
Cloud & DevOps: Docker, GitLab CI/CD (Continuous Integration/Continuous Deployment), GitHub Actions, DigitalOcean, AWS (Amazon Web Services), Proxmox, NGINX, Linux (Ubuntu/Arch)
Practices: Release management, code review, mentoring, technical interviewing, WCAG accessibility audits
Testing: TestCafe, automation and regression testing

Education

Self-taught software engineer. CS50x, Introduction to Computer Science (Harvard University, 2025), Introduction to Web Accessibility, WCAG (W3Cx, 2021).